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Isolation of chromosomes from *Pisum sativum* L. hairy root cultures and their analysis by flow cytometry

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Abstract

Continuously growing hairy-root cultures of pea (*Pisum sativum* L.) line L-84 have been induced by transformation of seedlings by *Agrobacterium rhizogenes* A4-24. The cultures were then used for cell cycle synchronisation in vitro and for preparation of chromosome suspensions. Various concentrations of hydroxyurea (0.5–2.0 mM) and timing of amiprophos-methyl (APM, 2–4 h) treatment have been tested to achieve the highest mitotic synchrony, and to accumulate the highest number of synchronised cells in metaphase, respectively. Optimal conditions (2 mM hydroxyurea for 18 h, 10 μM APM for 2 h applied 6 h after a release from hydroxyurea block) resulted in a high frequency of cells in metaphase (43%). The synchronised root tips were fixed in formaldehyde and chromosomes were released into a lysis buffer by mechanical homogenisation. The chromosomes in suspension showed a well preserved morphology, and could be used for flow cytometric analysis. While only two chromosomes (5 and 7) were discriminated in a standard karyotype, four chromosomes (3, 5, 6 and 7) could be clearly discriminated in the line L-84 which contains a stable reciprocal translocation between chromosomes 3 and 6. The establishment of hairy root cultures and the development of an efficient chromosome isolation procedure from a translocation line L-84 represent a basic prerequisite for subsequent flow-sorting of selected chromosomes of pea. © 1998 Elsevier Science Ireland Ltd. All rights reserved.

Keywords: Cell cycle synchronisation; Chromosome isolation; Flow cytometry; Flow karyotype; Hairy roots; *Pisum sativum* L.

1. Introduction

Flow-cytometric analysis of mitotic chromosomes is a technique allowing measurements of DNA content and AT/GC ratio in individual

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chromosome types in human, as well as in some animals and plants [1–3]. Chromosomes which differ in either of these parameters can be separated by flow sorting and resulting pure fractions of individual chromosome types can be used for subsequent molecular analysis. In plants, they have been employed for physical gene mapping by PCR [4], construction of chromosome-derived DNA libraries [5–7] and isolation of RFLP markers [6].

The availability of high-quality suspensions of intact metaphase chromosomes is a prerequisite for achieving high purity of flow-sorted chromosome fractions. Procedures for preparation of such chromosome suspensions have been established for several plant species (for review see [2,3]). A variety of tissues have been used for chromosome isolation, including cell suspensions in vitro [8,9,6,5], leaf mesophyll protoplasts [10,11], and root tip meristems [12–15]. Compared to cell cultures which are characterised by the frequent occurrence of chromosomal aberrations [16,17], the use of root tips offers the advantages of karyological stability and facile manipulation. Cell division in root tip meristems can also be efficiently synchronised, which substantially increases the yield of isolated chromosomes.

For successful sorting of chromosomes from a given species, two key requirements have to be fulfilled: (1) The tissue suitable for chromosome isolation and the optimal conditions for cell cycle synchronisation in this tissue have to be determined in order to obtain sufficient amounts of chromosomes for analysis; and (2) The chromosomes of the species of interest have to differ in their size (DNA content) so that they can be distinguished by flow-cytometric analysis. In pea (*Pisum sativum* L., $2n = 14$), the procedure for chromosome isolation from synchronised root tips of young seedlings has been developed by Gualberti et al. [13]. However, flow-cytometric analysis of standard (wild type) pea karyotype revealed that due to small differences in chromosome sizes only two chromosomes could be partially discriminated and none of them could be sorted in sufficient purity [13]. This is a common situation in plants and one of the main obstacles in the application of chromosome flow-sorting for genome analysis in many crop species. As shown for field bean (*Vicia faba* L.), this obstacle can be circumvented by using lines with reconstructed karyotypes which possess defined chromosomal translocations [18,4,7]. Since several lines with reconstructed karyotypes have also been described in pea [19,20], we attempted to use one of them, L-84 [19], for chromosome isolation and flow-cytometric analysis. The line L-84 carries a translocation between chromosomes 3 and 6 which makes them the smallest and the largest chromosomes in the karyotype, respectively. Because of the limited number of available seeds and the problems often associated with generative propagation of lines with reconstructed karyotypes, we focused on development of procedure for chromosome isolation from hairy-roots cultures. These cultures can be obtained by plant transformation by *Agrobacterium rhizogenes*, which induces differentiation of roots from transformed tissues [21,22]. The root cultures can be cultivated in vitro for years and are easily propagated. Since they represent a differentiated tissue they are genetically stable and can be used for chromosome isolation, as was shown for *Melandrium album* [23].

In this paper, we report on the development of a synchronisation procedure for pea hairy root cultures and subsequent isolation of metaphase chromosomes from these cultures. The chromosomes obtained from the line L-84 were analysed by flow-cytometry in order to measure the DNA content in individual chromosome types of this reconstructed karyotype. These data together with the optimised synchronisation protocol should serve as a basis for flow-sorting of pea chromosomes in the near future.

2. Material and methods

2.1. Plant material

Pea (*P. sativum* L., $2n = 14$) cultivar Ctirad with a standard (wild type) karyotype and a pea line L-84 containing a reciprocal translocation

between chromosomes 3 and 6 [19] were used for all experiments. Seeds of Ctirad were obtained from plant breeding station SEMO (Smržice, Czech Republic) and translocation line L-84 from Dr S. Lucretti (ENEA, Cassacia, Italy). To obtain continuously growing hairy root cultures from the line L-84, hypocotyles of 5 days-old seedlings germinated in vitro on Murashige–Skoog medium [24] were wounded by sterile needle and inoculated with a liquid culture of *A. rhizogenes* strain A4-24. After 2–3 weeks of cultivation, hairy roots emerging from the sites of inoculation were excised and placed on solid B5 [25] medium modified according to Tomes [26] and supplemented with ticarcilin for elimination of bacteria. Hairy root cultures were grown in glass petri dishes at 25°C in the dark and transferred to fresh B5 medium once a month.

2.2. Cell cycle synchronisation

Roots in rapidly growing hairy root cultures were cut 1–2 cm from the tip and transferred to Erlenmeyer flasks containing hydroxyurea at various concentrations (0.5, 1.0, 1.5, 2.0 mM) in liquid B5 medium. The roots were then treated for 18 h to accumulate cycling cells at the G_1/S interface. After this treatment, root tips were washed twice in hydroxyurea-free B5 medium and cultured in hydroxyurea-free medium for 10 h. All incubations were performed on a shaker at 25°C in the dark. Samples of roots were taken at 0.5- or 1-h intervals after the removal from the hydroxyurea block.

2.3. Accumulation of cells at metaphase

A phosphoric amide herbicide amiprophos-methyl (APM) was used at a concentration of 10 μM to block dividing cells at metaphase. The same concentration was used for pea seedlings [13] and our preliminary experiments showed that this concentration was optimal also for hairy roots. The compound was applied 6, 7 or 8 h after the release from the hydroxyurea-containing B5 medium for total of 2, 3 or 4 h. After the treatment, samples of roots were taken to evaluate the frequency of cells at metaphase.

2.4. Cytological analysis

Samples of roots were fixed in ethanol:acetic acid (3:1) overnight and stored in 70% ethanol. Feulgen-stained squash preparations were then prepared and used to analyse the frequency of cells in mitosis (mitotic index) or the frequency of cells in metaphase (metaphase index). From each treatment 10 roots were analysed, and 500–1000 cells were counted in each root meristem. All treatments were replicated three times and mean values were determined. Feulgen-stained slides were used also to determine relative length of individual chromosomes. In this case, 20 complete metaphases were analysed in each genotype. All slides were analysed with an Olympus BX-60 microscope.

2.5. Preparation of suspensions of intact chromosomes

A procedure originally developed for isolation of field bean chromosomes [12] has been used with some modifications. Briefly, synchronised root tips were washed in distilled water and fixed in 3% (v/v) formaldehyde made in Tris buffer (10 mM Tris–HCl, 10 mM Na_2EDTA, 100 mM NaCl, pH 7.5) for 25, 30 or 35 min at 5°C and finally washed three times in the Tris buffer for 20 min. Then, 300 meristems (ca. 1 mm long) of fixed root tips were cut and transferred to 1 ml of lysis buffer LB01 [27] into which mitotic chromosomes were released using a mechanical homogeniser (Polytron PT 1200). The speed of the homogeniser and the time of homogenisation was varied from 5000 to 20 000 rpm and from 5 to 30 s, respectively, to select optimal parameters. The homogenate was filtered through a 50 μm nylon mesh to remove large tissue fragments and cellular debris. Isolated chromosomes were stored at 4°C in Falcon 2054 tubes prior to analysis. Chromosome suspensions from a pea cultivar Ctirad with a standard (wild type) karyotype were prepared as described previously [13].

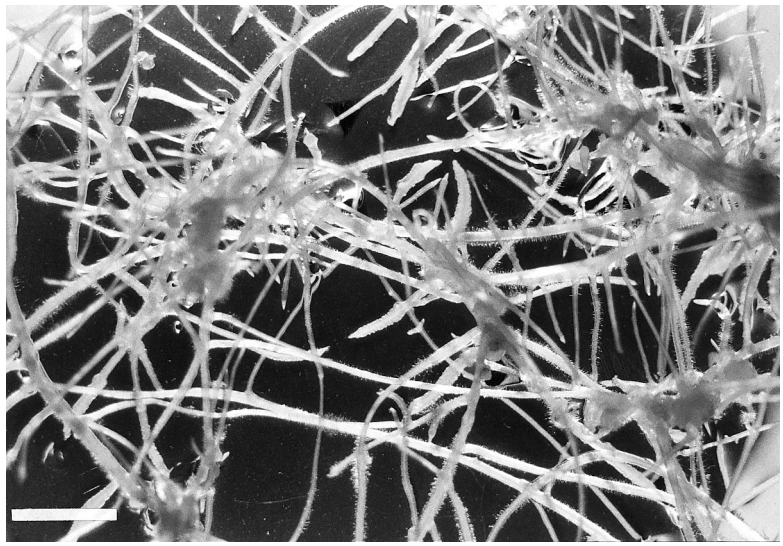


Fig. 1. Hairy root culture of pea (*P. sativum*) L. line L-84 4 weeks after transfer to a fresh medium. The culture was induced by transformation with *A. rhizogenes* strain A4-24. Bar = 1 cm.

2.6. Flow cytometric analysis

Chromosome suspensions were stained with DAPI (4,′6-diamino-2-phenylindole) at 5 μM and fluorescence intensity was analysed using a Partec PAS II flow cytometer (Partec GmbH, Münster). The instrument was equipped with a high pressure mercury arc lamp HBO 100W/L and with a standard optical filter set for analysis of DAPI fluorescence. Theoretical flow karyotypes were modelled on the basis of relative chromosome length using a KARYOSTAR computer software [28].

3. Results and discussion

3.1. Hairy root cultures

Hairy root cultures of pea line L-84 were obtained by transformation of young seedlings by *A. rhizogenes* A4-24. Several independently transformed lines of hairy roots were selected based on their high rate of growth on hormone-free solid medium. These cultures maintained their phenotype of continuously growing and occasionally branching hairy roots (Fig. 1) for more than 3 years of cultivation. Karyological analysis revealed no chromosomal abnormalities in the root tip meristems (data not shown).

3.2. Cell cycle synchronisation

Mitotic index in the root-tip meristems of hairy root cultures varied between 4 and 6%. In order to increase the proportion of cells simultaneously undergoing cell division, the procedure for cell cycle synchronisation using hydroxyurea has been optimised. As expected, the ability of hydroxyurea to block the cell cycle depended on its concentration. Lower concentrations (0.5 or 1.0 mM) of hydroxyurea did not prevent the cells from entering the S-phase which was reflected by only a slight increase of the mitotic index after the removal from hydroxyurea (Fig. 2A). On the other hand, the use of hydroxyurea at 1.5 or 2.0 mM resulted in considerable increase in mitotic activity 6 h after release from the block. Maximal mitotic indexes (34%) were observed 7.5 and 8 h after the release from the hydroxyurea at 1.5 and 2 mM, respectively (Fig. 2B).

The results indicated a higher sensitivity of hairy root meristem cells to hydroxyurea compared to seedlings. While a peak of mitotic activity in pea seedlings was observed already 4 h after

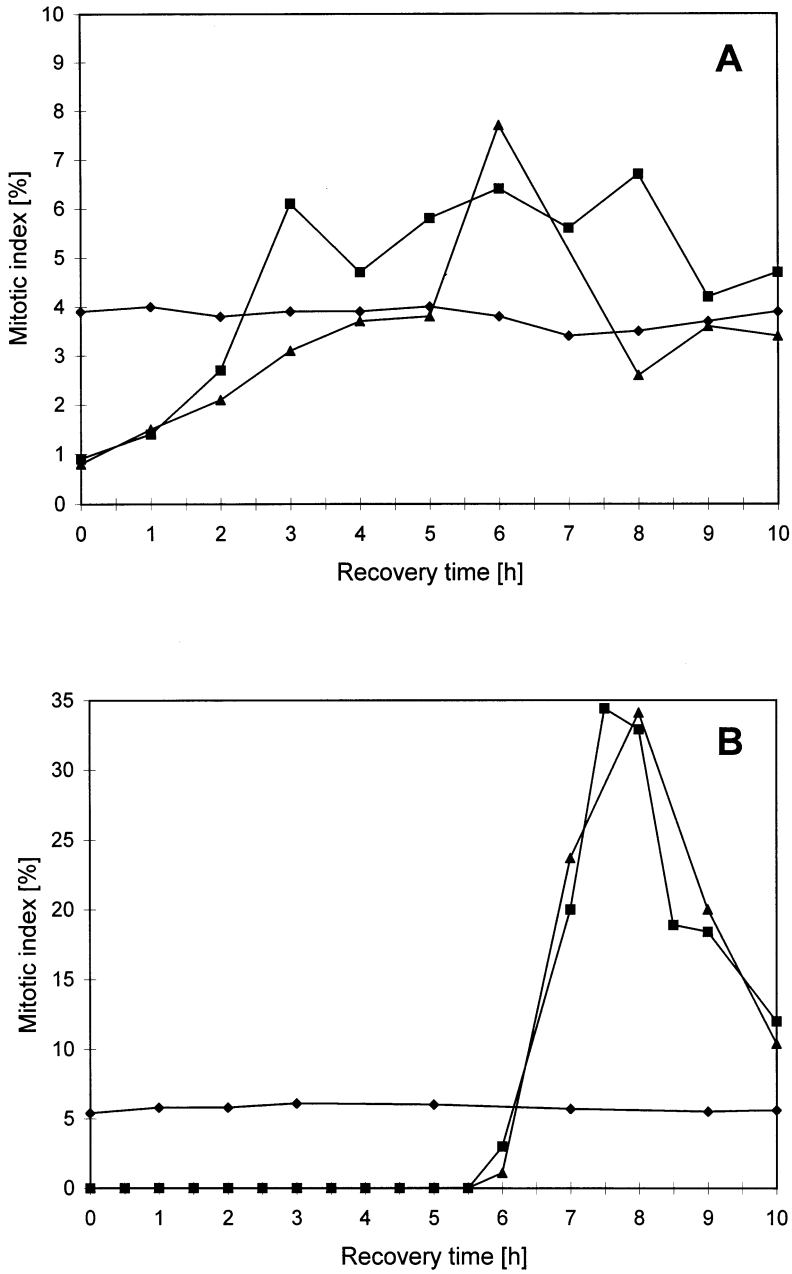


Fig. 2. Mitotic activity observed in root tip meristems of pea hairy root culture observed during a recovery from the 18 h hydroxyurea block. (A) hydroxyurea was used at 0 mM (◆), 0.5 mM (■) or 1.0 mM (▲); (B) hydroxyurea was used at 0 mM (◆), 1.5 mM (■) or 2.0 mM (▲).

the removal from 1.25 mM hydroxyurea [13], in hairy roots, dividing cells were not observed until 6 h layer and the mitotic activity peaked only 7 h after the removal from 1.5 mM hydroxyurea. In both cases, the hydroxyurea incubation lasted 18 h. The data of Gualberti [13] suggest that in their case the cells entered the S-phase already before hydroxyurea removal. It is interesting to note,

that a synchronous release of cells from the block before the end of hydroxyurea treatment was observed also in other species [29]. On the other hand, our results (Fig. 2) clearly show that hairy root cells were effectively blocked from entering the S-phase with hydroxyurea at similar concentration. Higher sensitivity of hairy root cultures to hydroxyurea has already been observed in the field bean (*V. faba*) [30] and thus may represent a common phenomenon.

3.3. Accumulation of cells at metaphase

Accumulation of cells at metaphase was achieved by treating synchronised roots with 10 μM amiprophos-methyl (APM). Based on the results described above, the APM was applied for various time periods starting 6–8 h after removal of hydroxyurea from the medium. The data describing the relationship of cell frequency at mitosis (mitotic index) with APM treatment are summarised in Fig. 3. Both the mitotic index and the percentage of cells blocked at metaphase depended on the start point and the duration of the APM treatment. The efficiency of APM to block cells at metaphase was very high and cells at anaphase or telophase were observed at very low frequencies (ranging from 0 to 1%) after the treatment. Prolonging the APM treatment increased the metaphase frequency up to 54% but also increased the occurrence of cells with single-chromatid chromosomes (c-anaphases) (data not shown). Thus, we have chosen a 2-h treatment with 10 μM APM applied 6 h after releasing the cells from the hydroxyurea block as optimal (Fig. 4). Using this treatment, more than 43% of cells at metaphase were routinely observed.

3.4. Preparation of chromosome suspensions

Chromosomes were released from formaldehyde-fixed root tips using a mechanical homogeniser. The morphology and yield of isolated chromosomes, as determined by microscopic observations and flow-cytometric analysis, depended on the extent of fixation with formaldehyde and on the extent of homogenisation (data not shown). Fixation for 35 min in 3% formaldehyde followed by homogenisation at 12 000 rpm for 15 s gave the best results. In this case most of the isolated chromosomes were intact and the suspension contained the least amount of chromosome clumps. On average, 4.5×10^5 chromosomes were isolated from 300 root meristems. This corresponds to a yield of 1.5×10^3 chromosomes from one root tip. Compared to the use of pea seedlings [13] the chromosome yield per root tip was lower, which was clearly due to the smaller size of hairy root meristems. However, this can be compensated by using larger number of roots which can be easily multiplied in vitro.

3.5. Flow cytometric analysis

Theoretical flow-karyotypes were modelled by the KARYOSTAR computer program [28] based on relative chromosome lengths obtained in this study (Table 1). The expected distribution of chromosome peaks on histograms of relative DNA content (flow karyotypes) is shown in Fig. 5. In the case of a standard karyotype, only two peaks corresponding to individual chromosomes (5 and 7) could be discriminated. A composite peak representing chromosomes 1 and 2 and the major composite peak representing chromosomes 3, 4 and 6 were observed. However, the predicted discrimination of chromosomes 5 and 7 was not very good with the two peaks being positioned close to each other and to the main composite peak (Fig. 5A). Compared to the standard karyotype, the theoretical flow karyotype of the line L-84 predicted very good discrimination of four chromosomes (3, 5, 6 and 7) (Fig. 5B).

With some minor differences, the experimental histograms obtained after flow-cytometric analysis of DAPI-stained chromosomes confirmed theoretical models (Fig. 6). The flow karyotype of a standard line consisted of one dominant composite peak corresponding to chromosomes 3, 4 and 6. In contrast to the model, the peak representing chromosomes 1 and 2 almost overlapped with the main composite peak of chromosomes 3, 4 and 6 thus forming a poorly resolved shoulder.

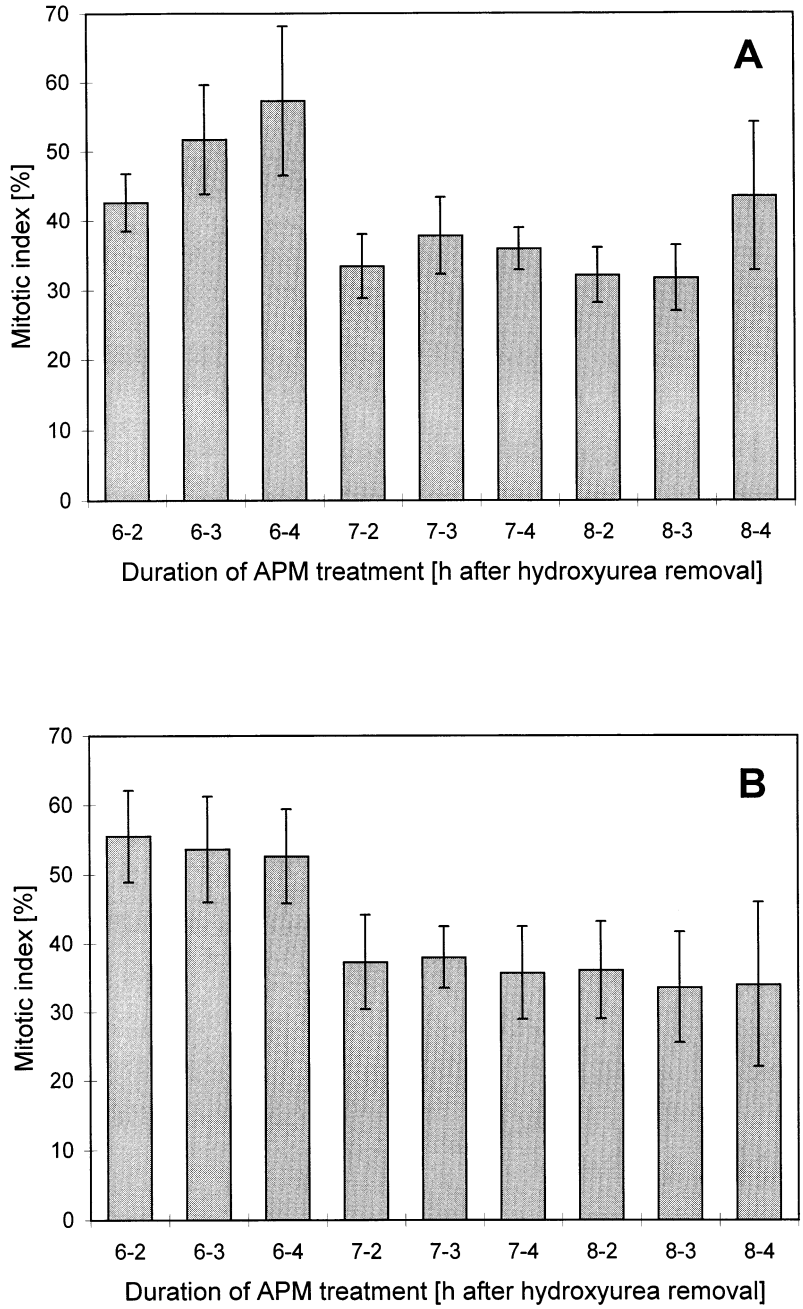


Fig. 3. Mitotic activity and frequency of various phases of mitosis observed in root tip meristems of pea hairy root culture x h after the 18 h hydroxyurea block at 1.5 (A) or 2.0 mM (B) followed by APM treatment for y h at 10 μM. The annotation (x–y) indicates, respectively, treatment delay and duration.

Furthermore, a partially discriminated peak corresponding to chromosome 7 and a clearly resolved peak corresponding to chromosome 5 were observed. These peaks were better separated from the main peak and also from each other than predicted by the model. Similar differences be-

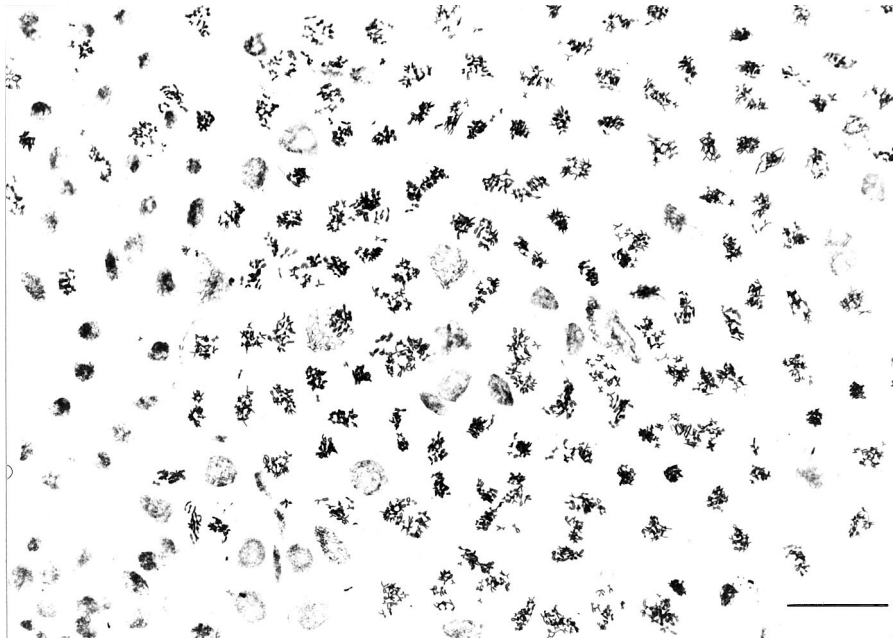


Fig. 4. Accumulation of metaphases in synchronized root tip meristems of pea hairy root culture induced by a combined use of hydroxyurea (18 h, 2 mM) and APM (2 h, 10 μM). Bar = 60 μm.

tween theoretical and experimental flow karyotypes were observed also in the field bean [12,18] and could be explained by differences in chromatin accessibility for DAPI and/or by differences in AT/GC ratio between individual chromosomes.

The experimental flow karyotype of a translocation line L-84 consisted of four clearly discriminated peaks corresponding to chromosomes 3, 5, 6 and 7 and a composite peak corresponding to chromosomes 1, 2 and 4 (Fig. 6B). Compared to the theoretical model, peaks of chromosomes 1, 2 an 4 were closer to each other thus forming a narrower composite peak. In both cases (standard and L-84 karyotype), the resolution of flow karyotypes was relatively high with coefficients of variation of DNA (chromosome) peaks ranging from 1.1 to 2.6%.

This study demonstrated that hairy root cultures represent a convenient system for long-term propagation of a pea line with a reconstructed karyotype and that these cultures are suitable for preparation of suspensions of intact chromosomes. While only two chromosomes could be discriminated using a standard karyotype of pea, the use of translocation line L-84 permitted discrimination of four chromosomes (i.e. more than half of the chromosome complement) using flow cytometry. Sorting of the chromosome types which are not possible to obtain from the line L-84 can be achieved by application of the same approach to other available translocation lines. Moreover, there is still an unexplored possibility to use the transformation step to insert a large enough DNA sequence which could serve as a chromosome-specific marker for labelling some of the morphologically similar chromosomes. We have already shown that DNA sequences of suitable size could be fluorescently labelled on chromosomes in suspension by primed in situ DNA labelling en suspension (PRINSES) [31] and that this procedure permitted sorting of otherwise indistinguishable chromosomes [32]. These results indicate that it will be possible to obtain pure fractions of individual chromosomes in pea by flow sorting which will aid in gene isolation and mapping in this crop.

Table 1
Relative chromosome lengths in pea (*P. sativum* L.) lines with a standard (wild type) and reconstructed karyotype (L-84)[a]

Karyotype	Chromosome number						
	1	2	3	4	5	6	7
Standard	12.65 (0.3)	12.04 (0.49)	14.06 (0.51)	14.39 (0.7)	16.86 (0.76)	14.39 (0.92)	15.68 (0.5)
Reconstructed	12.95 (1.0)	11.95 (1.01)	8.1 (0.46)	14.06 (0.5)	17.2 (0.31)	20.15 (0.36)	15.56 (0.43)

[a] The value in brackets means standard deviation.

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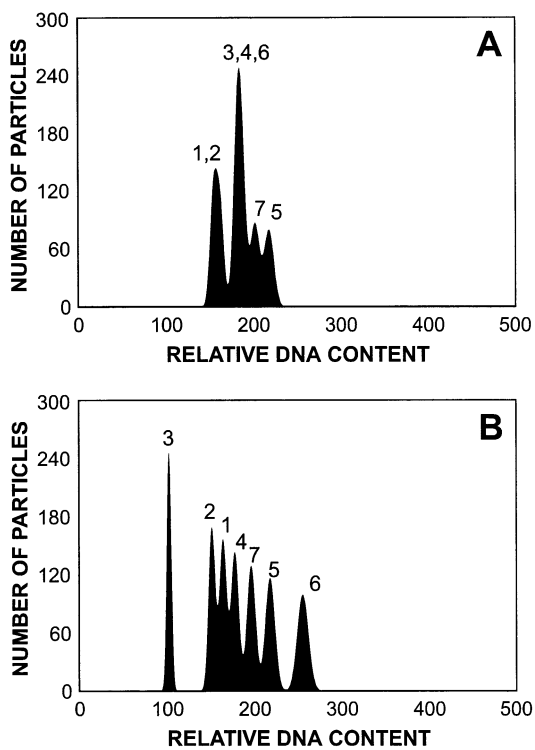


Fig. 5. Theoretical flow karyotypes modelled for pea (*P. sativum* L., $2n = 14$). (A) cultivar Ctirad with a standard (wild type) karyotype; (B) line L-84 with a reconstructed karyotype carrying a reciprocal translocation between chromosomes 3 and 6. The models were calculated based on relative chromosome length (c.f. Table 1) and 2.5% coefficient of variation of DNA peaks. Numbers indicate chromosomes to which individual peaks were assigned.

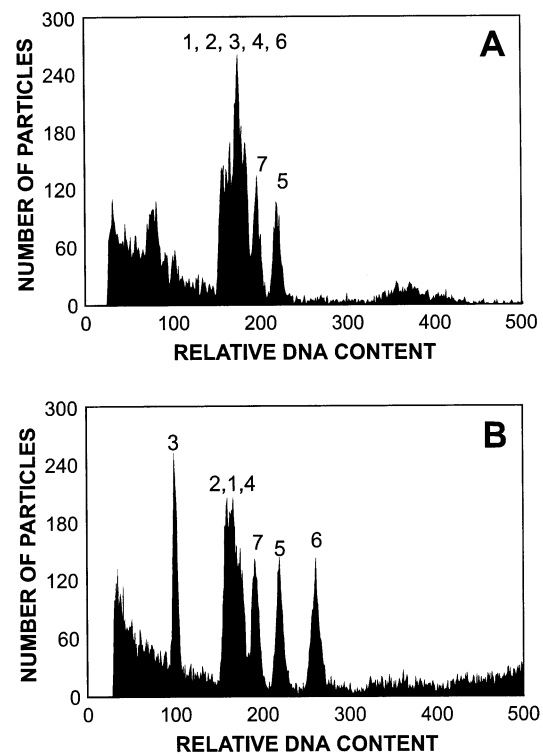


Fig. 6. Typical flow karyotypes obtained after the analysis of chromosomes isolated from pea (*P. sativum* L., $2n = 14$) cultivar Ctirad with a standard (wildtype) karyotype (A), and from a line L-84 with a reconstructed karyotype carrying a reciprocal translocation between chromosomes 3 and 6 (B). Chromosomes were stained by DAPI and their relative DNA content was analysed by flow cytometry. The numbers indicate chromosomes to which individual peaks were assigned.

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